

DGR LAB1 STP v1.5.2 - Jorge Parra ITIT 13104



Universidad La Salle Chihuahua

Diseño y Gestión de Redes

LAB 1 — Spanning Tree

Alumno: Jorge Parra

Semestre: Noveno

Clave: ITI2025-09

Profesor: Ing. Raul Adrian Chaparro Guzmán

Fecha: 18 de agosto de 2026

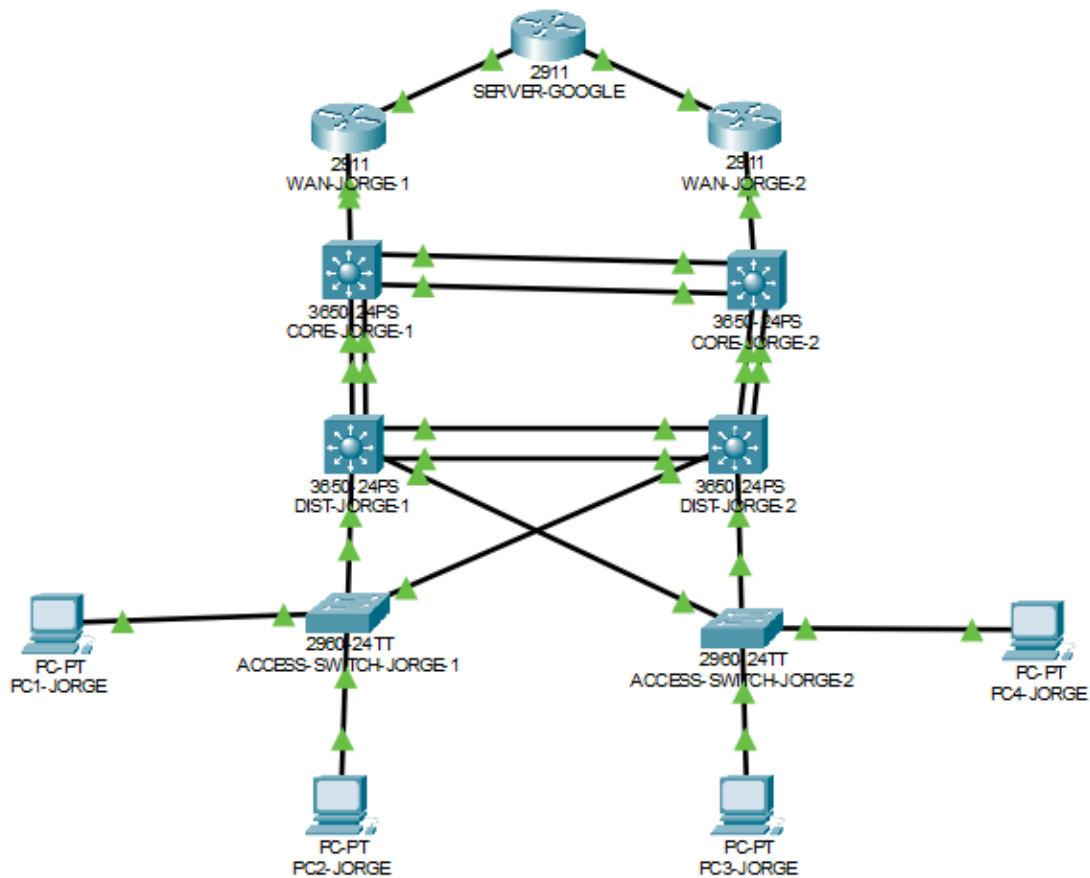
1. Introducción

En esta práctica se implementó y verificó el funcionamiento de Spanning Tree Protocol (STP) en una topología jerárquica redundante. Se configuró Rapid-PVST en switches de acceso, distribución y núcleo, con los switches CORE como referencia para la elección del Root Bridge. También se analizaron los roles y estados de puertos para identificar enlaces en **Forwarding** y enlaces **bloqueados** para evitar bucles de Capa 2.

La verificación se realizó principalmente con `show spanning-tree vlan`, revisando modo, prioridades, roles y estados por VLAN.

2. Evidencia y explicación

2.1 Topología



2.2 Capturas (comandos / resultados)

ACCESS-SWITCH-JORGE1

```

ACCESS-SWITCH-JORGE1>
ACCESS-SWITCH-JORGE1>ena
ACCESS-SWITCH-JORGE1>enable
ACCESS-SWITCH-JORGE1#show spanning-tree vlan 90
VLAN0090
  Spanning tree enabled protocol rstp
    Root ID      Priority    24666
                Address      000A.4155.47B3
                Cost         23
                Port         23 (FastEthernet0/23)
                Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

    Bridge ID    Priority    32858  (priority 32768 sys-id-ext 90)
                Address      0001.9624.54BA
                Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
                Aging Time   20

Interface      Role Sts Cost      Prio.Nbr Type
-----
Fa0/1          Desg FWD 19        128.1    P2p
Fa0/23         Root FWD 19        128.23   P2p
Fa0/24         Altn BLK 19        128.24   P2p
ACCESS-SWITCH-JORGE1#

```

ACCESS-SWITCH-JORGE2

```

ACCESS-SWITCH-JORGE2#show spanning-tree vlan 90
VLAN0090
  Spanning tree enabled protocol rstp
    Root ID      Priority    24666
                Address      000A.4155.47B3
                Cost         23
                Port         24 (FastEthernet0/24)
                Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

    Bridge ID    Priority    32858  (priority 32768 sys-id-ext 90)
                Address      00E0.8F76.A3DB
                Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
                Aging Time   20

Interface      Role Sts Cost      Prio.Nbr Type
-----
Fa0/23         Altn BLK 19        128.23   P2p
Fa0/24         Root FWD 19        128.24   P2p
ACCESS-SWITCH-JORGE2#

```

DIST-SWITCH-JORGE1

```

DIST-SWITCH-JORGE1>ena
DIST-SWITCH-JORGE1>enable
DIST-SWITCH-JORGE1#show spanning-tree vlan 90
VLAN0090
  Spanning tree enabled protocol ieee
    Root ID    Priority    24666
              Address    000A.4155.47B3
              Cost        4
              Port        24(GigabitEthernet1/0/24)
              Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

    Bridge ID  Priority    32858  (priority 32768 sys-id-ext 90)
              Address    00D0.BA15.B675
              Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time  20

Interface                Role Sts Cost        Prio.Nbr Type
-----
Gi1/0/1                  Desg FWD 19          128.1    P2p
Gi1/0/2                  Desg FWD 19          128.2    P2p
Gi1/0/10                 Desg FWD 4          128.10   P2p
Gi1/0/11                 Desg FWD 4          128.11   P2p
Gi1/0/23                 Altn BLK 4          128.23   P2p
Gi1/0/24                 Root FWD 4          128.24   P2p

```

DIST-SWITCH-JORGE1#

DIST-SWITCH-JORGE2

```

DIST-SWITCH-JORGE2#
DIST-SWITCH-JORGE2#show spanning-tree vlan 90
VLAN0090
  Spanning tree enabled protocol rstp
    Root ID    Priority    24666
              Address    000A.4155.47B3
              Cost        8
              Port        24(GigabitEthernet1/0/24)
              Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

    Bridge ID  Priority    32858  (priority 32768 sys-id-ext 90)
              Address    0001.64D5.7288
              Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time  20

Interface                Role Sts Cost        Prio.Nbr Type
-----
Gi1/0/2                  Desg FWD 19          128.2    P2p
Gi1/0/1                  Desg FWD 19          128.1    P2p
Gi1/0/10                 Altn BLK 4          128.10   P2p
Gi1/0/11                 Altn BLK 4          128.11   P2p
Gi1/0/23                 Altn BLK 4          128.23   P2p
Gi1/0/24                 Root FWD 4          128.24   P2p

```

DIST-SWITCH-JORGE2#

CORE-SWITCH-JORGE1

```

CORE-SWITCH-JORGE1>
CORE-SWITCH-JORGE1>enab
CORE-SWITCH-JORGE1>enable
CORE-SWITCH-JORGE1#show spanning-tree vlan 90
VLAN0090
  Spanning tree enabled protocol rstp
  Root ID    Priority    24666
             Address     000A.4155.47B3
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    24666 (priority 24576 sys-id-ext 90)
             Address     000A.4155.47B3
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Gi1/0/1	Desg	FWD	4	128.1	P2p
Gi1/0/2	Desg	FWD	4	128.2	P2p
Gi1/0/20	Desg	FWD	4	128.20	P2p
Gi1/0/21	Desg	FWD	4	128.21	P2p

```

CORE-SWITCH-JORGE1#show spanning-tree vlan 180
VLAN0180
  Spanning tree enabled protocol rstp
  Root ID    Priority    24756
             Address     000A.4155.47B3
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    24756 (priority 24576 sys-id-ext 180)
             Address     000A.4155.47B3
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Gi1/0/1	Desg	FWD	4	128.1	P2p
Gi1/0/2	Desg	FWD	4	128.2	P2p

CORE-SWITCH-JORGE2

```

CORE-SWITCH-JORGE2#
CORE-SWITCH-JORGE2#show spanning-tree vlan 90
VLAN0090
  Spanning tree enabled protocol rstp
    Root ID    Priority    24666
              Address     000A.4155.47B3
              Cost        4
              Port        20 (GigabitEthernet1/0/20)
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

    Bridge ID  Priority    28762  (priority 28672 sys-id-ext 90)
              Address     00E0.A3AE.D935
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time   20

Interface      Role Sts Cost      Prio.Nbr Type
-----
Gi1/0/1        Desg FWD 4         128.1    P2p
Gi1/0/2        Desg FWD 4         128.2    P2p
Gi1/0/21       Altn BLK 4         128.21   P2p
Gi1/0/20       Root FWD 4         128.20   P2p

CORE-SWITCH-JORGE2#show spanning-tree vlan 180
VLAN0180
  Spanning tree enabled protocol rstp
    Root ID    Priority    24756
              Address     000A.4155.47B3
              Cost        4
              Port        20 (GigabitEthernet1/0/20)
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

    Bridge ID  Priority    28852  (priority 28672 sys-id-ext 180)
              Address     00E0.A3AE.D935
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time   20

Interface      Role Sts Cost      Prio.Nbr Type
-----
Gi1/0/1        Desg FWD 4         128.1    P2p
Gi1/0/2        Desg FWD 4         128.2    P2p

```

2.3 Glosario rápido (STP)

- **FWD:** el puerto reenvía tráfico.
- **BLK:** el puerto está bloqueado por STP.
- **Root:** puerto con el mejor camino hacia el Root Bridge.
- **Desg (Designated):** puerto designado como mejor salida en un segmento.
- **Altn (Alternate):** camino alternativo (normalmente bloqueado).

3. Análisis por switch (por VLAN)

Nota: Si en una captura aparece **LRN** (Learning), espera unos segundos y vuelve a ejecutar el comando hasta que cambie a **FWD/BLK**.

3.1 CORE-SWITCH-JORGE1 (Root Bridge)

VLAN 90, 180, 270 y 360

- **Modo:** Rapid-PVST (RSTP).
- **Rol del switch:** Root Bridge.
- **Puertos hacia otros switches (Gi1/0/1, Gi1/0/2, Gi1/0/20, Gi1/0/21):**
 - **Desg / FWD** (no hay puertos bloqueados en este switch).

3.2 CORE-SWITCH-JORGE2 (secundario)

VLAN 90

- **No es Root Bridge** (Root: CORE-SWITCH-JORGE1).
- **Gi1/0/20:** Root / FWD (camino principal al Root Bridge).
- **Gi1/0/21:** Altn / **BLK** (camino redundante bloqueado).
- **Gi1/0/1 y Gi1/0/2:** Desg / FWD.

VLAN 180, 270 y 360

Puerto	Estado	Rol
Gi1/0/1	FWD	Desg
Gi1/0/2	FWD	Desg
Gi1/0/20	FWD	Root
Gi1/0/21	BLK	Altn

3.3 DIST-SWITCH-JORGE1

VLAN 90, 180, 270 y 360

- **Root Bridge:** CORE-SWITCH-JORGE1.
- **Gi1/0/24:** Root / FWD (camino principal).
- **Gi1/0/23:** Altn / **BLK** (camino redundante).
- **Gi1/0/1, Gi1/0/2, Gi1/0/10, Gi1/0/11:** Desg / FWD.

3.4 DIST-SWITCH-JORGE2

VLAN 90

- **Gi1/0/24:** Root / FWD.
- **Gi1/0/10, Gi1/0/11, Gi1/0/23:** Altn / **BLK.**
- **Gi1/0/1 y Gi1/0/2:** Desg / FWD.

3.5 ACCESS-SWITCH-JORGE1

VLAN 90

- **Fa0/23:** Root / FWD.
- **Fa0/24:** Altn / **BLK.**
- **Fa0/1:** Desg / FWD (host).

VLAN 180

- **Fa0/2:** Desg / FWD.
- **Fa0/23:** Root / FWD.
- **Fa0/24:** Altn / **BLK.**

VLAN 270 y 360

- **Fa0/23:** Root / FWD.
- **Fa0/24:** Altn / **BLK.**

3.6 ACCESS-SWITCH-JORGE2

VLAN 90

- **Fa0/24:** Root / FWD.
- **Fa0/23:** Altn / **BLK.**

VLAN 180

- **Fa0/24:** Root / FWD.
- **Fa0/23:** Altn / **BLK.**

VLAN 270

- **Fa0/24:** Root / FWD.

- **Fa0/23:** Altn / **BLK**.
- **Fa0/1:** Desg / FWD.

VLAN 360

- **Fa0/24:** Root / FWD.
- **Fa0/23:** Altn / **BLK**.
- **Fa0/2:** Desg / FWD.

4. Tabla

Estados STP por switch e interfaz

RSTP - resumen de forwarding, blocking y roles de puerto

SWITCH	INTERFAZ	STP MODE / PRIORIDAD	ESTADO	ROL
CORE1	Gi1/0/1	RSTP - 24666	FWD	Desg
	Gi1/0/2	RSTP - 24666	FWD	Desg
	Gi1/0/20	RSTP - 24666	FWD	Desg
	Gi1/0/21	RSTP - 24666	FWD	Desg
CORE2	Gi1/0/1	RSTP - 28762	FWD	Desg
	Gi1/0/2	RSTP - 28762	FWD	Desg
	Gi1/0/20	RSTP - 28762	FWD	Root
	Gi1/0/21	RSTP - 28762	BLK	Altn
DIST1	Gi1/0/1	RSTP - 32858	FWD	Desg
	Gi1/0/2	RSTP - 32858	FWD	Desg
	Gi1/0/23	RSTP - 32858	BLK	Altn
	Gi1/0/24	RSTP - 32858	FWD	Root
DIST2	Gi1/0/1	RSTP - 32858	FWD	Desg
	Gi1/0/2	RSTP - 32858	FWD	Desg
	Gi1/0/23	RSTP - 32858	BLK	Altn
	Gi1/0/24	RSTP - 32858	FWD	Root
ACCESS1	Fa0/23	RSTP - 32858	FWD	Root
	Fa0/24	RSTP - 32858	BLK	Altn
ACCESS2	Fa0/23	RSTP - 32858	BLK	Altn
	Fa0/24	RSTP - 32858	FWD	Root

5. Conclusión

La implementación de Rapid-PVST permitió mantener una topología redundante sin bucles de Capa 2. Mediante la elección del Root Bridge y la asignación de roles **Root/Designated/Alternate**, STP determinó qué enlaces permanecerían en **Forwarding** y cuáles debían bloquearse. Las verificaciones con `show spanning-tree` `vlan` confirmaron el funcionamiento del protocolo y la distribución de estados y roles en switches CORE, Distribution y Access.